

Partial Progress on Strategyproofness versus Group-Strategyproofness for Leontief Budget Aggregation

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Abstract

Brandt, Greger, Segal-Halevi, and Suksompong characterize the Nash-product rule for Leontief budget aggregation as the unique continuous mechanism satisfying group-strategyproofness and core fair share. They ask whether group-strategyproofness can be weakened to ordinary strategyproofness. This note does *not* resolve that question in full. It records partial progress and useful reductions: the answer is affirmative when there are at most two agents, while the two-alternative case follows from the one-dimensional uniform-phantom characterization of Brandt et al. The two-agent result is a short consequence of two facts: core fair share implies Pareto efficiency for Leontief utilities, and for two agents strategyproofness plus Pareto efficiency implies group-strategyproofness. We also give an exact finite inequality test for core fair share and a small rational example showing that a purely local endpoint argument cannot settle the remaining case $n \geq 3, m \geq 3$.

1 Introduction and status

Let Δ^m be the simplex of distributions over m alternatives. Under Leontief utilities, agent i reports a peak $p_i \in \Delta^m$ and evaluates an outcome $q \in \Delta^m$ by the smallest preserved fraction of her positive peak coordinates. Brandt, Greger, Segal-Halevi, and Suksompong [1] show that the Nash-product rule is the unique continuous mechanism satisfying group-strategyproofness and core fair share. They explicitly ask whether the same characterization remains true if group-strategyproofness is weakened to ordinary strategyproofness.

The present note is partial progress only. It proves the conjectured implication in the boundary cases $n \leq 2$ and $m = 2$, but it does not settle the genuinely multidimensional many-agent case $n \geq 3, m \geq 3$. The $n \leq 2$ argument is included because it is short and robust: for two agents, once core fair share gives Pareto efficiency, no joint manipulation by both agents is possible. The $m = 2$ argument is a direct consequence of the one-dimensional characterization in [1]. Finally, Section 6 gives a concrete obstruction to a tempting local proof strategy for the remaining case.

2 Model and known characterization

Let $N = [n]$ be the set of agents and $M = [m]$ the set of alternatives. Outcomes are distributions

$$q \in \Delta^m := \left\{ q \in \mathbb{R}_{\geq 0}^m : \sum_{j \in M} q_j = 1 \right\}.$$

Agent i reports a peak $p_i \in \Delta^m$. Write

$$S_i := \text{supp}(p_i) = \{j \in M : p_{i,j} > 0\}.$$

The Leontief utility of agent i at profile $P = (p_1, \dots, p_n)$ is

$$u_i^P(q) = u_{p_i}(q) := \min_{j \in S_i} \frac{q_j}{p_{i,j}}.$$

Equivalently, $u_i^P(q)$ is the largest scalar t such that $q_j \geq tp_{i,j}$ for every $j \in S_i$.

A mechanism is a map $f : (\Delta^m)^n \rightarrow \Delta^m$. It is *strategyproof* if, for every profile P , every agent i , and every report $p'_i \in \Delta^m$,

$$u_{p_i}(f(p_i, P_{-i})) \geq u_{p_i}(f(p'_i, P_{-i})).$$

It is *group-strategyproof* if there is no nonempty coalition $G \subseteq N$ and no joint misreport P'_G such that, writing

$$q = f(P), \quad q' = f(P'_G, P_{-G}),$$

one has

$$u_i^P(q') \geq u_i^P(q) \quad \text{for every } i \in G, \quad u_i^P(q') > u_i^P(q) \quad \text{for at least one } i \in G.$$

Thus strategyproofness is the restriction of group-strategyproofness to singleton coalitions.

A distribution q is *Pareto efficient* at profile P if there is no $r \in \Delta^m$ such that

$$u_i^P(r) \geq u_i^P(q) \quad \text{for every } i \in N, \quad u_i^P(r) > u_i^P(q) \quad \text{for at least one } i \in N.$$

The Nash-product rule is

$$\text{NASH}(P) = \arg \max_{q \in \Delta^m} \prod_{i=1}^n u_i^P(q),$$

which is single-valued for the Leontief domain considered here; see [1].

We will use the following theorem of Brandt et al. as a black box.

Theorem 1 (Brandt–Greger–Segal–Halevi–Suksompong). *For Leontief utilities, the Nash-product rule is the only continuous mechanism satisfying group-strategyproofness and core fair share.*

3 An exact finite test for core fair share

We use the weak-core version of core fair share: a coalition blocks only if all its members can be made strictly better off. Fix a profile P and an outcome q . For a nonempty coalition $G \subseteq N$, put

$$\alpha_G := \frac{|G|}{n}.$$

The coalition G *blocks* q if there is an internal distribution $y \in \Delta^m$ such that, for every outside distribution $z \in \Delta^m$,

$$u_i^P(\alpha_G y + (1 - \alpha_G)z) > u_i^P(q) \quad \text{for every } i \in G.$$

The outcome q satisfies *core fair share* at P if no nonempty coalition blocks it.

For a coalition G , define

$$A_G(q, P) := \sum_{j \in M} \max_{i \in G} (u_i^P(q) p_{i,j}),$$

where $u_i^P(q) p_{i,j}$ is interpreted as 0 when $p_{i,j} = 0$.

Proposition 2 (Exact core-fair-share inequalities). *Fix a Leontief profile P , an outcome q , and a nonempty coalition $G \subseteq N$. Then G blocks q if and only if*

$$A_G(q, P) < \frac{|G|}{n}.$$

Consequently, q satisfies core fair share at P if and only if

$$A_G(q, P) \geq \frac{|G|}{n} \quad \text{for every nonempty } G \subseteq N.$$

Proof. Let $\alpha = |G|/n$ and abbreviate $u_i = u_i^P(q)$. First suppose that $A_G(q, P) < \alpha$. Put

$$a_j := \max_{i \in G} u_i p_{i,j}.$$

Since $\sum_j a_j < \alpha$, choose numbers $c_j > a_j$ for all $j \in M$ with $\sum_j c_j = \alpha$, and define $y \in \Delta^m$ by $y_j = c_j/\alpha$. Then, for every outside distribution $z \in \Delta^m$ and every $i \in G$,

$$\alpha y_j + (1 - \alpha) z_j \geq c_j > a_j \geq u_i p_{i,j} \quad (j \in S_i).$$

Thus every positive peak coordinate of each $i \in G$ receives strictly more than an u_i -multiple of her peak, and therefore

$$u_i^P(\alpha y + (1 - \alpha) z) > u_i^P(q) \quad (i \in G).$$

So G blocks q .

Conversely, suppose G blocks q via $y \in \Delta^m$. If $m = 1$, then $q = y = z = (1)$ and every agent's utility is already 1, so no blocking is possible; also $A_G(q, P) = 1 \geq \alpha$. Hence assume $m \geq 2$.

For each coordinate j , if $\alpha < 1$ choose an outside distribution z with $z_j = 0$; if $\alpha = 1$, the outside term is irrelevant. Since the blocking inequality must hold for every outside distribution, for every $i \in G$ and every $j \in S_i$ we get

$$\alpha y_j > u_i p_{i,j}.$$

Therefore, for every coordinate j with $a_j > 0$,

$$\alpha y_j > a_j.$$

Summing over these coordinates gives

$$\alpha = \sum_j \alpha y_j \geq \sum_{j: a_j > 0} \alpha y_j > \sum_{j: a_j > 0} a_j = A_G(q, P).$$

Thus $A_G(q, P) < \alpha$. □

Corollary 3 (Core fair share implies Pareto efficiency). *For Leontief utilities, every core-fair-share outcome is Pareto efficient.*

Proof. Apply Theorem 2 to the grand coalition $G = N$. If q satisfies core fair share, then

$$\sum_j \max_i u_i^P(q) p_{i,j} \geq 1.$$

For every i and j , the definition of Leontief utility gives

$$q_j \geq u_i^P(q) p_{i,j}.$$

Hence

$$1 = \sum_j q_j \geq \sum_j \max_i u_i^P(q) p_{i,j} \geq 1,$$

so equality holds throughout and therefore

$$q_j = \max_i u_i^P(q) p_{i,j} \quad (j \in M). \quad (1)$$

Now let $r \in \Delta^m$ weakly improve every agent at P , i.e. $u_i^P(r) \geq u_i^P(q)$ for all i . Then, for every i and j ,

$$r_j \geq u_i^P(r) p_{i,j} \geq u_i^P(q) p_{i,j}.$$

Taking the maximum over i and using (1), we get $r_j \geq q_j$ for every j . Since both r and q have total mass 1, this forces $r = q$. Thus no strict Pareto improvement exists. $\square$

4 The case of at most two agents

The two-agent case follows from a general observation that has nothing specifically to do with Leontief utilities.

Proposition 4. *If $n \leq 2$, strategyproofness plus Pareto efficiency implies group-strategyproofness.*

Proof. For $n = 1$, group-strategyproofness and strategyproofness are the same condition. Let $n = 2$. A profitable manipulation by a singleton coalition is exactly a profitable unilateral manipulation, so it is ruled out by strategyproofness. The only other nonempty coalition is the grand coalition N . Suppose the two agents could jointly misreport from P to some P' and thereby move the outcome from $q = f(P)$ to $q' = f(P')$ while weakly improving both true utilities and strictly improving at least one. Then q' would be a Pareto improvement over q at the true profile P , contradicting Pareto efficiency of $q = f(P)$. Hence no coalition can profitably manipulate. $\square$

Theorem 5 (Affirmative answer for $n \leq 2$). *For $n \leq 2$ and arbitrary m , every continuous, strategyproof, core-fair-share mechanism for Leontief utilities coincides with the Nash-product rule.*

Proof. Let f be continuous, strategyproof, and core-fair-share. By Theorem 3, every outcome of f is Pareto efficient. By Theorem 4, f is group-strategyproof. The conclusion now follows from Theorem 1. $\square$

Remark 6. *This is only a boundary-case result. The argument uses the special fact that for $n = 2$ the only non-singleton coalition is the grand coalition, whose profitable deviations are ruled out by Pareto efficiency. For $n \geq 3$, a coalition can exclude some agents, so Pareto efficiency of the truthful outcome does not by itself rule out a coalition manipulation.*

5 The case of two alternatives

For completeness we record the two-alternative case. This is not new beyond the one-dimensional characterization of Brandt et al.; it is included to delimit the remaining open case.

A profile is *single-minded* if every peak is degenerate, i.e. each p_i is a unit vector e_j . A mechanism satisfies *proportionality* if, on every single-minded profile,

$$f(P) = \frac{1}{n} \sum_{i=1}^n p_i.$$

Lemma 7. *For Leontief utilities, core fair share implies proportionality.*

Proof. Fix a single-minded profile and an outcome q satisfying core fair share. For each alternative j , let

$$G_j := \{i \in N : p_i = e_j\}.$$

If G_j is nonempty, then each agent in G_j has utility q_j at outcome q , and

$$A_{G_j}(q, P) = q_j.$$

By Theorem 2, core fair share gives

$$q_j \geq \frac{|G_j|}{n}.$$

This also holds trivially when $G_j = \emptyset$. Summing over j yields equality for every j , because

$$\sum_j q_j = 1 = \sum_j \frac{|G_j|}{n}.$$

Thus $q_j = |G_j|/n$ for every alternative j , which is precisely proportionality. $\square$

Theorem 8 (Affirmative answer for $m = 2$). *For arbitrary n and $m = 2$, every continuous, strategyproof, core-fair-share mechanism for Leontief utilities coincides with the Nash-product rule.*

Proof. When $m = 2$, the simplex can be identified with the interval $[0, 1]$ by writing an outcome as $(1 - x, x)$. Leontief utilities then form a subdomain of single-peaked preferences on $[0, 1]$: an agent with peak x_i has utility

$$u_{x_i}(x) = \begin{cases} 1 - x, & x_i = 0, \\ x, & x_i = 1, \\ \min \left\{ \frac{1 - x}{1 - x_i}, \frac{x}{x_i} \right\}, & 0 < x_i < 1. \end{cases}$$

By Theorem 7, core fair share implies proportionality. Brandt et al.'s one-dimensional characterization says that, for $m = 2$ and any single-peaked subdomain with one admissible utility for every peak, the only continuous strategyproof mechanism satisfying proportionality is the uniform phantom rule. Their corollary for Leontief utilities states that this uniform phantom rule is exactly the Nash-product rule. Hence $f = \text{NASH}$. $\square$

6 A local obstruction for the remaining case

The remaining case is $n \geq 3$ and $m \geq 3$. The following example does not construct a strategyproof mechanism and therefore is not a counterexample to the open question. Its purpose is narrower: it shows that a proof cannot rely only on endpoint core fair share and on the two individual strategyproofness inequalities of a single moving agent.

Example 9 (Endpoint CFS does not force local nonbossiness). *Let $n = m = 3$ and consider*

$$P = \left(\left(\frac{9}{11}, \frac{1}{11}, \frac{1}{11} \right), (1, 0, 0), \left(\frac{1}{20}, \frac{19}{20}, 0 \right) \right),$$

with outcome

$$q = \left(\frac{3}{5}, \frac{1}{3}, \frac{1}{15} \right).$$

Agent 1's critical alternatives at (P, q) are 1 and 3. The corresponding report supported on those critical alternatives is

$$p_1^* = \left(\frac{9}{10}, 0, \frac{1}{10} \right),$$

and we write

$$P^* = \left(\left(\frac{9}{10}, 0, \frac{1}{10} \right), (1, 0, 0), \left(\frac{1}{20}, \frac{19}{20}, 0 \right) \right).$$

Finally set

$$q' = \left(\frac{37}{60}, \frac{19}{60}, \frac{1}{15} \right).$$

Then $q \neq q'$, but the changing agent is indifferent in both directions:

$$u_{p_1}(q) = u_{p_1}(q') = \frac{11}{15}, \quad u_{p_1^*}(q) = u_{p_1^*}(q') = \frac{2}{3}.$$

Moreover, q satisfies core fair share at P , and q' satisfies core fair share at P^* .

Proof. The equalities for agent 1 follow by direct computation:

$$u_{p_1}(q) = \min \left\{ \frac{3/5}{9/11}, \frac{1/3}{1/11}, \frac{1/15}{1/11} \right\} = \frac{11}{15},$$

$$u_{p_1}(q') = \min \left\{ \frac{37/60}{9/11}, \frac{19/60}{1/11}, \frac{1/15}{1/11} \right\} = \frac{11}{15},$$

and

$$u_{p_1^*}(q) = \min \left\{ \frac{3/5}{9/10}, \frac{1/15}{1/10} \right\} = \frac{2}{3}, \quad u_{p_1^*}(q') = \min \left\{ \frac{37/60}{9/10}, \frac{1/15}{1/10} \right\} = \frac{2}{3}.$$

It remains to check core fair share using Theorem 2. At (P, q) , the utility vector is

$$\left(\frac{11}{15}, \frac{3}{5}, \frac{20}{57} \right),$$

and the scaled demand vectors $u_i^P(q)p_i$ are

$$\left(\frac{3}{5}, \frac{1}{15}, \frac{1}{15} \right), \quad \left(\frac{3}{5}, 0, 0 \right), \quad \left(\frac{1}{57}, \frac{1}{3}, 0 \right).$$

The singleton coalition values are

$$A_{\{1\}} = \frac{11}{15}, \quad A_{\{2\}} = \frac{3}{5}, \quad A_{\{3\}} = \frac{20}{57},$$

all at least $1/3$. The pair values are

$$A_{\{1,2\}} = \frac{11}{15}, \quad A_{\{1,3\}} = 1, \quad A_{\{2,3\}} = \frac{14}{15},$$

all at least $2/3$, and the grand-coalition value is $A_N = 1$. Thus q is core-fair-share at P .

At (P^*, q') , the utility vector is

$$\left(\frac{2}{3}, \frac{37}{60}, \frac{1}{3}\right),$$

and the scaled demand vectors are

$$\left(\frac{3}{5}, 0, \frac{1}{15}\right), \quad \left(\frac{37}{60}, 0, 0\right), \quad \left(\frac{1}{60}, \frac{19}{60}, 0\right).$$

The singleton coalition values are

$$A_{\{1\}} = \frac{2}{3}, \quad A_{\{2\}} = \frac{37}{60}, \quad A_{\{3\}} = \frac{1}{3},$$

all at least $1/3$. The pair values are

$$A_{\{1,2\}} = \frac{41}{60}, \quad A_{\{1,3\}} = \frac{59}{60}, \quad A_{\{2,3\}} = \frac{14}{15},$$

all at least $2/3$, and again $A_N = 1$. Hence q' is core-fair-share at P^* . $\square$

Remark 10. *At the true profile P , moving from q to q' leaves agent 1 indifferent, raises agent 2's utility from $3/5$ to $37/60$, and lowers agent 3's utility from $20/57$ to $1/3$. Thus the outcome change transfers utility among nonmoving agents while preserving the moving agent's two individual incentive inequalities. This is exactly the phenomenon that the two-agent proof cannot accommodate when $n \geq 3$.*

7 Conclusion

The open question of Brandt et al. remains open in the main case $n \geq 3, m \geq 3$: this note does not prove that strategyproofness and core fair share characterize the Nash-product rule there, and it does not provide a counterexample. The positive results proved here are limited to $n \leq 2$, where strategyproofness plus efficiency already implies group-strategyproofness, and to $m = 2$, where the issue is one-dimensional and follows from the uniform-phantom characterization. The exact core-fair-share inequalities in Theorem 2 may be useful for further work, while Theorem 9 shows that any proof for the remaining case must use genuinely global incentive constraints rather than only local endpoint information.

References

- [1] Felix Brandt, Matthias Greger, Erel Segal-Halevi, and Warut Suksompong. *Optimal Budget Aggregation with Star-Shaped Preference Domains*. Mathematics of Operations Research, 2025. arXiv:2402.15904; DOI: 10.1287/moor.2024.0723.